

Remote Monitoring of Epilepsy Patients with Wearable Device and Smartphone

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ABSTRACT

Emerging mobile health (mHealth) and eHealth technologies offer new opportunities for remote monitoring and interventions for individuals with mental health and neurological disorders. Traditional hospital methods are not suitable for long-term epilepsy seizure monitoring in naturalistic ambulatory environments. A study was conducted using a wearable device, smartphone, and remote data collection platform to explore the feasibility of ambulatory seizure detection. 27 patients were recruited and monitored over six months in the UK and Germany. Active and passive data were collected, indicating the potential for seizure detection in at-home and ambulatory settings. Collected data completion rate and quality shows feasibility of remote data collection. Initial results suggest that smartphones and wearable devices could significantly improve care for patients with epilepsy by detecting and potentially preempting epileptic seizures.

CCS CONCEPTS

• **Human-centered computing** → **Smartphones; Mobile devices**; • **Applied computing** → **Health informatics**.

KEYWORDS

mHealth, Wearable Devices, Smartphones, Data Collection Platform, Mental Health, Epilepsy, Seizures

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1 INTRODUCTION

There has been an enormous increase in the capability to monitor individuals via smartphones and wearable devices during the last decade, with a growing range of parameters offered by such technologies for continuous measurement [12]. The RADAR-base platform [1, 13] enables study design and set up, active and passive remote data collection. It provides secure data transmission and scalable data storage, management and access solutions. The platform offers two leading apps: the active remote monitoring app (aRMT) and the passive remote monitoring app (pRMT).

Epilepsy is a neurological condition characterized by a person's tendency to have epileptic seizures. The global prevalence of epilepsy is between 4-10 per 1000 people. Those with epilepsy have a reduced life expectancy; people with symptomatic epilepsy have a life expectancy 18 years shorter [5]. We hypothesize that wearable devices designed for consumers could offer ongoing seizure detection. This capability might lead to better-informed decisions regarding the administration of anti-epileptic drugs, offering a more objective assessment of an individual's epilepsy status.

While hospital monitoring systems such as Video/EEG/ ECG are utilized for home monitoring, they are unsuitable for long-term seizure detection in home-based settings. To address this limitation, we leveraged the RADAR-base platform to investigate the viability of three wearable devices for detecting seizures in ambulatory settings. Data collection occurred over a maximum period of 14

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days per patient. An in-hospital study [3, 14] was conducted using various wearable devices to assess the feasibility of seizure detection. The Empatica E4 device successfully detected seizures, which laid the groundwork for an ambulatory study. In this follow-up study, 27 epilepsy patients were recruited and monitored over six months using Empatica E4 wearable devices. This monitoring was complemented by data collection from smartphone sensors and health assessment questionnaires.

The main aims of the study were: (1) To determine the usability and acceptability of RMT for providing real-time, objective, multidimensional indications of clinical state in individuals with epilepsy at home. (2) To assess the correspondence between patient-reported seizure events (via aRMT) and data collected from the pRMT. (3) To determine whether multiparametric RMT data collected from individuals with epilepsy can prospectively estimate variations in seizure occurrence and other outcomes, including seizure frequency, quality of life (QoL), and comorbidities.

The initial results demonstrate the feasibility of remote data collection using wearable devices from patients with epilepsy.

2 RELATED WORK

Several relevant studies and mHealth platforms for remote monitoring in mental health are discussed here [12]. Detection of seizures using non-EEG wearable devices has been well studied over the last decade [6]. However, the performance of the proposed models could have been better, particularly in terms of specificity, although some studies do show some promising results for seizures with a significant motor component [2]. Additionally, few studies have been conducted outside of an in-patient environment, so the performance of these models in the real world is unknown. The field has moved towards detection using multiple modalities to improve the accuracy of models that only use a single sensor type, usually an accelerometer [6]. A few studies have used using multiple sensors. Poh et al. used electrodermal activity (EDA) and an accelerometer and showed increased GTCS detection performance when using both as opposed to only acceleration in 7 patients [11].

Heldberg et al. also used EDA and an accelerometer, looking at convulsive and non-convulsive seizures in 8 patients [9]. Other studies have examined the combination of acceleration and ECG-derived cardiac features [8, 15]. Using multiple sensors does not always uniformly lead to better performance; Milosevic et al. report improved seizure detection but lower specificity when using both accelerometers and electromyography [10].

In this study, we utilized multiple sensors from commercially available wearable devices and smartphone sensors to confirm the accuracy of detected seizures. Additionally, patients completed assessment questionnaires through the active monitoring app to further validate the findings.

3 METHODS AND PROCEDURES

Figure 1 shows the system overview. Empatica E4 is streaming data to the pRMT app via bluetooth, aRMT and pRMT are streaming data to the RADAR-base platform. Figure 2 shows the screenshots of apps and wearable device.

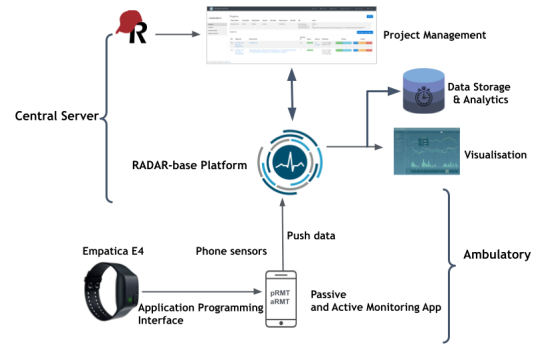


Figure 1: System Overview: Empatica E4 and Smartphone sensors are streaming data to RADAR-base platform. Ambulatory section shows patients can wear devices and stream data from home settings.

3.1 Study Population

This multicenter, prospective, observational cohort study involved 27 individuals diagnosed with epilepsy. Participants were asked to download several RMT apps and use a wearable device for up to 6 months of follow-up. The study was nonrandomized and noninterventional.

3.2 Active Remote Monitoring App

A variety of key variables for this study were gathered through RADAR-base platform active Remote Monitoring (aRMT) app. aRMT is available both on Google Play and iOS App Store. App prompts participants to complete validated questionnaires and assessments according to the study schedule. Assessments include Psychiatric comorbidities, Seizure occurrence and precipitants, Experience sampling method, Speech, Cognitive.

3.3 Passive Remote Monitoring App

Passive Remote Monitoring app (pRMT) is an android based app available on Google Play, participants will be guided in downloading the apps, they operates in the background without needing further interaction. Utilizing the sensors found in all modern smartphones, pRMT gathers data on ambient noise, ambient light, GPS location, Bluetooth connectivity, and battery life. To ensure privacy, the GPS location data was randomized, offering relative location information rather than exact coordinates.

3.4 Empatica E4 Device

Participants were requested to wear the Empatica E4 device, a CE-marked wristband, throughout the entire 6-month duration of the study. The device requires a few hours of charging daily. To accommodate this, participants were provided with training, accessories, and a second E4 device to wear during charging periods. This wearable device features sensors for measuring PPG and derived interbeat intervals, 3D acceleration, body temperature, and EDA.



Figure 2: RADAR-base platform Active and Passive app integrated with Empatica E4 device.

3.5 RADAR-base Platform

To facilitate the wearables and smartphone data remote collection at scale and digital biomarkers development, the RADAR-base platform was released under the open-source Apache 2 licence in January 2018 [16]. RADAR-base is composed of an Apache Kafka-based back-end deployed onto Kubernetes infrastructure and two mobile apps. The cross-platform (Android, iOS) Cordova Active Remote Monitoring App (aRMT) for active monitoring of participants, requiring conscious action (e.g. questionnaires, audio questions, timed tests etc.), and the Passive Remote Monitoring App (pRMT), a native Android app for passive monitoring of participants via phone, wearable devices and IoT.

3.6 Data Collection for Epilepsy Seizure Detection

Hospital in-patient participants were recruited for the RADAR-EPI substudy of RADAR-CNS study [4] as part of an otherwise typical stay at the Clinical Neurophysiology Department at Kings' College Hospital (KCH), London, UK or the Epilepsy Center at the University Hospital of Freiburg, Germany (UKFLR) [14]. Patients were monitored by a video-EEG and seizures are annotated by clinicians as part of the routine clinical assessment of their seizures. This provides ready-made source of gold-standard labels for use in developing wearable-based seizure detection methods. In parallel, each patient wore 1 to 3 of the study wearable devices; the Empatica E4 wristband, Faros 180, Biovotion VSM1, or an offline IMEC device. The IMEC device records data offline, which was routinely transferred to the RADAR-base storage server. The other devices sent data to the RADAR-base platform via a Bluetooth-paired android device through the passive RMT app. Figure 3 shows the In-Hospital study design and overview. Aim of the In-Hospital study was to choose appropriate wearable device and test data collection platform for the present ambulatory study.

3.7 Statistical and Analysis Plan

The initial analysis of the epilepsy data requires a different approach. Seizures are typically short and sparse, so the primary challenge is to be able to detect the relatively short periods of ictal activity between the much more common segments of interictal time, while keeping the false positive rate at an acceptable level. Initial focus will be on

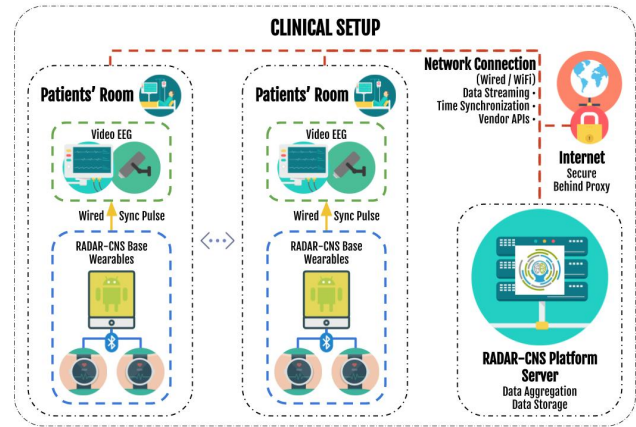


Figure 3: In-Hospital Study for selection of device and feasibility of apps, Wearable devices are streaming data to RADAR-base platform in parallel to video-EEG.

the offline detection of generalised and focal seizures with a motor component, particularly those with tonic or clonic movements, using a combination of the available signals. The combination of the different signal modalities should improve the estimation accuracy of a single relevant parameter, and also allow the analysis using multiple physiologically-relevant parameters, enabling a panoramic view of the patient's status.

Because a few participants have had a large number of seizures ($n > 15$) during their in-patient stay, there is an opportunity to measure the performance of an individualised seizure detection algorithm as successive seizures are added to the model. This has application to current ambulatory study, in which it is important to know how much data is required for an adequately accurate model.

Firstly, we will follow an analytical pipeline similar to those in prior seizure detection work, extracting features from the EDA, accelerometer, and heart rate signals, and classifying the ictal period of focal motor seizures using a support vector machine. Additionally, we will try and determine feature importance with the intention of elucidating performance gain from including additional signal modalities.

4 RESULTS

Figure 4 illustrates the data completion and patient engagement levels for the active app, passive app, and wearable device. Some patients managed to stream over 90% of the data, demonstrating the potential for remote data collection through wearables and smartphones. Missing data was primarily due to battery outages, Bluetooth connectivity issues between the Empatica E4 and the phone, and participant engagement issues with the active app. These challenges can be addressed through increased patient engagement with study managers and enhanced technology training. Table 1 presents the data completion rates across both study sites.

A tonic EDA response during the post-ictal period has been noted elsewhere[7], and often occurs within this study collected dataset. An example is given in Figure 5, showing an Empatica E4

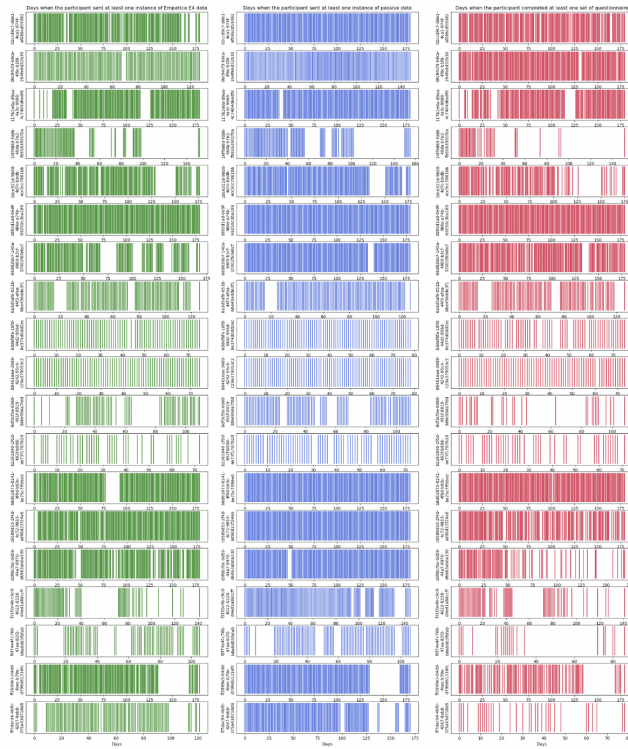


Figure 4: Data streamed by patients over the period of their follow-up. a) Passive data, b) Active data, c) Wearable device data

recording of acceleration and EDA. The convulsive seizure at 16:10 BST is followed by a large increase in skin conductance, with a peak at 16:12. There are other tonic peaks in the EDA, but they do not coincide with a seizure-like accelerometer trace. Equally, there is no evidence accelerometer traces with repetitive or otherwise confusable characteristics in the inter-ictal period being succeeded by an EDA response. Although not totally consistent across all participants and all seizures, it is a general pattern that illustrates the potential to use multiple modalities for increased specificity.

5 CONCLUSION AND FUTURE WORKS

Results suggest that wearable technology can effectively gather critical health data outside of traditional clinical settings, potentially improving continuous monitoring and management of epilepsy. The success of this approach highlights its promise for enhancing patient care through more frequent and detailed data collection, which can lead to better-informed treatment decisions and improved patient outcomes.

Future studies will involve further analysis of the collected data based on the plan outlined in section 3.6. Additionally, the new version of the Empatica E4, known as Embrace Plus, along with other newly released devices that boast improved data collection performance, will be considered for use.

Table 1: Data Completion Rate

Active and Passive Apps	KCL	UKFLR	Overall
aRMT	49.10%	60.20%	54.90%
pRMT	63.70%	83.90%	74.40%
Empatica E4	53.20%	69.20%	61.60%

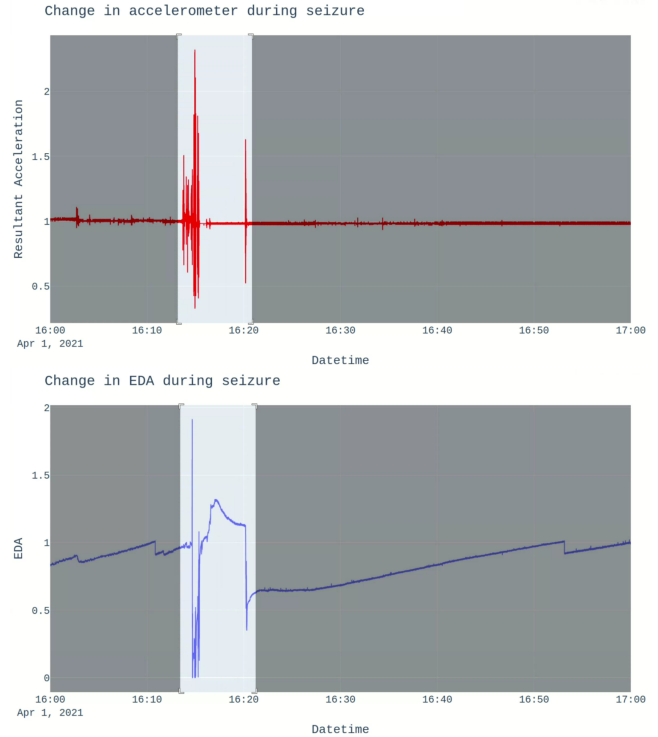


Figure 5: Data stream from a patient wearing an Empatica E4. The patient had a focal motor seizure at 16:10 (BST), corresponding to a burst of movement in the accelerometer (top), and subsequently followed by a peak in EDA (bottom). Other movements and peaks in EDA during the interictal periods do not follow the same pattern.

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REFERENCES

- [1] 2018. RADAR-base. <https://radar-base.org/>.
- [2] Sándor Beniczky, Isa Conradsen, Oliver Henning, Martin Fabricius, and Peter Wolf. 2018. Automated real-time detection of tonic-clonic seizures using a wearable EMG device. *Neurology* (2018). <https://doi.org/10.1212/WNL.0000000000004893>
- [3] Elisa Bruno, Andrea Biondi, Sebastian Böttcher, Gergely Vértés, Richard Dobson, Amos Folarin, Yatharth Ranjan, Zulqarnain Rashid, Nikolay Manyakov, Aki Rintala, Inez Myin-Germers, Sara Simblett, Til Wykes, Amanda Stoneman, Ann Little, Sarah Thorpe, Simon Lees, Andreas Schulze-Bonhage, and Mark Richardson. 2020. Remote Assessment of Disease and Relapse in Epilepsy: Protocol for a Multicenter Prospective Cohort Study. *JMIR Res Protoc* 9, 12 (16 Dec 2020), e21840. <https://doi.org/10.2196/21840>
- [4] RADAR-CNS consortium. 2016. RADAR-CNS: Remote Assessment of Disease and Relapse in Central Nervous System Disorders. <https://www.radar-cns.org/>.
- [5] Hanneke M. de Boer, Marco Mula, and Josemir W Sander. 2008. The global burden and stigma of epilepsy. *Epilepsy and Behavior* 12, 4 (2008), 540 – 546. <https://doi.org/10.1016/j.yebeh.2007.12.019> Current Views on Epilepsy and Behavior.
- [6] Anouk Van de Vel, Kris Cuppens, Bert Bonroy, Milica Milosevic, Katrien Jansen, Sabine Van Huffel, Bart Vanrumste, Patrick Cras, Lieven Lagae, and Berten Ceulemans. 2016. Non-EEG seizure detection systems and potential SUDEP prevention: State of the art: Review and update. *Seizure* (2016).
- [7] Onorati Francesco, Regalia Giulia, Caborni Chiara, Migliorini Matteo, Bender Daniel, Poh Ming-Zher, Frazier Cherise, Kovitch Thropp Eliana, Mynatt Elizabeth D., Bidwell Jonathan, Mai Roberto, LaFrance W. Curt, Blum Andrew S., Friedman Daniel, Loddenkemper Tobias, Mohammadpour-Touserani Fatemeh, Reinsberger Claus, Tognetti Simone, and Picard Rosalind W. [n.d.]. Multicenter clinical assessment of improved wearable multimodal convulsive seizure detectors. *Epilepsia* 58, 11 ([n.d.]), 1870–1879. <https://doi.org/10.1111/epi.13899> arXiv:<https://onlinelibrary.wiley.com/doi/pdf/10.1111/epi.13899>
- [8] F. Furbass, S. Kampusch, E. Kaniusas, J. Koren, S. Pirker, R. Hopfengärtner, H. Stefan, T. Kluge, and C. Baumgartner. 2017. Automatic multimodal detection for long-term seizure documentation in epilepsy. *Clinical Neurophysiology* 128, 8 (2017), 1466 – 1472. <https://doi.org/10.1016/j.clinph.2017.05.013>
- [9] B. E. Heldberg, T. Kautz, H. Leutheuser, R. Hopfengärtner, B. S. Kasper, and B. M. Eskofier. 2015. Using wearable sensors for semiology-independent seizure detection - towards ambulatory monitoring of epilepsy. In *2015 37th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, 5593–5596. <https://doi.org/10.1109/EMBC.2015.7319660>
- [10] M. Milošević, A. Van de Vel, B. Bonroy, B. Ceulemans, L. Lagae, B. Vanrumste, and S. V. Huffel. 2016. Automated Detection of Tonic-Clonic Seizures Using 3-D Accelerometry and Surface Electromyography in Pediatric Patients. *IEEE Journal of Biomedical and Health Informatics* 20, 5 (Sept 2016), 1333–1341. <https://doi.org/10.1109/JBHI.2015.2462079>
- [11] Poh Ming-Zher, Loddenkemper Tobias, Reinsberger Claus, Swenson Nicholas C., Goyal Shubhi, Sabtala Mangwe C., Madsen Joseph R., and Picard Rosalind W. [n.d.]. Convulsive seizure detection using a wrist-worn electrodermal activity and accelerometry biosensor. *Epilepsia* 53, 5 ([n.d.]), e93–e97. <https://doi.org/10.1111/j.1528-1167.2012.03444.x> arXiv:<https://onlinelibrary.wiley.com/doi/pdf/10.1111/j.1528-1167.2012.03444.x>
- [12] John A. Naslund, Lisa A. Marsch, Gregory J. McHugo, and Stephen J. Bartels. 2015. Emerging mHealth and eHealth interventions for serious mental illness: a review of the literature. *Journal of Mental Health* 24, 5 (2015), 321–332. <https://doi.org/10.3109/09638237.2015.1019054> arXiv:<https://doi.org/10.3109/09638237.2015.1019054> PMID: 26017625.
- [13] Yatharth Ranjan, Zulqarnain Rashid, Callum Stewart, Pauline Conde, Mark Begale, Denny Verbeeck, Sebastian Boettcher, Richard Dobson, and Amos Folarin. 2019. RADAR-Base: Open Source Mobile Health Platform for Collecting, Monitoring, and Analyzing Data Using Sensors, Wearables, and Mobile Devices. *JMIR Mhealth Uhealth* 7, 8 (01 Aug 2019), e11734. <https://doi.org/10.2196/11734>
- [14] Zulqarnain Rashid, Callum L. Stewart, Sebastian Böttcher, Yatharth Ranjan, Richard J.B Dobson, and Amos A. Folarin. 2018. RADAR-base: Epilepsy Case Study. In *Proceedings of the 2018 ACM International Joint Conference and 2018 International Symposium on Pervasive and Ubiquitous Computing and Wearable Computers* (Singapore, Singapore) (*UbiComp '18*). Association for Computing Machinery, New York, NY, USA, 227–230. <https://doi.org/10.1145/3267305.3267578>
- [15] Shivkumar Sabesan and Raman Sankar. 2015. Improving long-term management of epilepsy using a wearable multimodal seizure detection system. *Epilepsy & Behavior* 46 (2015), 56–57.
- [16] Rashid Z, Folarin AA, Zhang Y, Ranjan Y, Conde P, Sankesara H, Sun S, Stewart C, Laiou P, and Dobson RJ. [n.d.]. RADAR-base Platform: Digital phenotyping of mental and physical conditions through remotely collected wearables and smartphone data. *JMIR Mental Health* ([n.d.]). <https://doi.org/10.2196/51259>